

AP Computer Science A: The Complete Exam Guide

The 4-unit exam explained, a free-response playbook, Java traps to avoid, a month-by-month plan, and 10 practice questions with full explanations.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**3 hrs**

90 min multiple choice
+ 90 min free response

42+4

multiple-choice +
free-response questions

Java

typed in Bluebook,
fully digital exam

66–67%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) combines a multiple-choice section with four free-response questions where you write real Java. Both sections get 90 minutes, and the split is close to even.

55%

Section I · Multiple choice · 42 questions

45%

Section II · Free response · 4 questions

Exam structure

Section	Weight & timing	What you do
I · Multiple choice	55% of score 90 minutes 42 questions	Mostly stand-alone questions, plus 1–2 small sets of 2 questions that share a code segment. You read, trace and evaluate Java code. There is no penalty for wrong answers.
II · Free response	45% of score 90 minutes 4 questions	Write Java methods, a constructor and a full class. Q1 Methods & Control Structures · Q2 Class Design · Q3 Data Analysis with ArrayList · Q4 2D Array. Same four types every year.

FULLY DIGITAL

You type your code in Bluebook

Both sections are taken in College Board's Bluebook app. Free-response code is typed into the built-in editor and submitted automatically at the end. The **Java Quick Reference** (the library methods you may use) is available inside Bluebook.

THE CURRENT COURSE

Four units, 25 free-response points

The course is organized into 4 large units. Free-response questions total 25 points, Q1 Part B asks you to call **String** methods, and Q3 is about **ArrayList**. Older practice tests from the 10-unit course still help, but don't match exactly.

How to use this guide

Just starting?

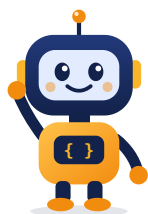
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 20 minutes, then use the final-four-weeks plan.



Pigo's tip: Download Bluebook early and open the AP CSA test preview. Typing Java in that editor (no autocomplete!) feels different from your usual IDE, so practice there before May.

SECTION I

02 The multiple-choice section

42 questions in 90 minutes gives you a little over 2 minutes per question. Most of that time goes into tracing loops and method calls, so practice tracing on paper under a timer.

42

questions, each with one correct answer

~2:10

minutes per question on average

55%

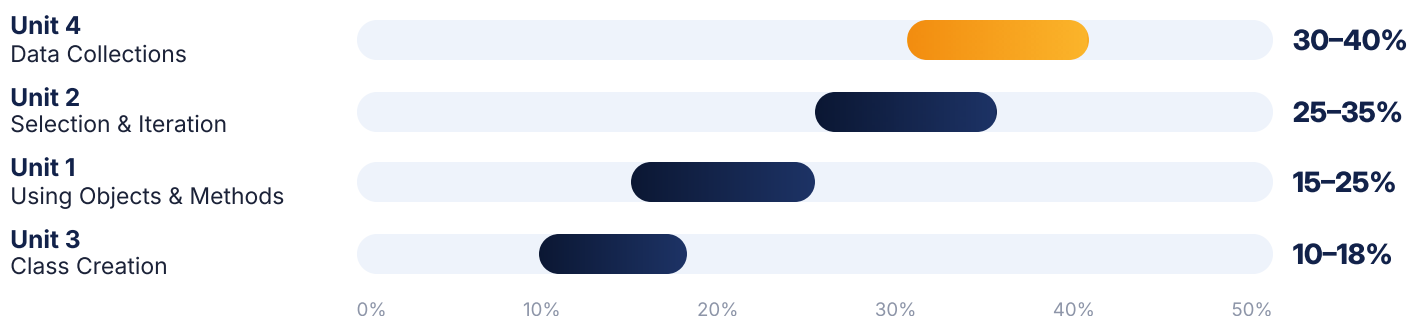
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the multiple-choice section. Bars show minimum to maximum.

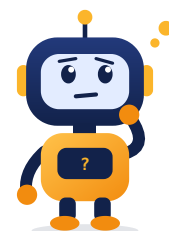


What this means for your study time: Units 2 and 4 together are well over half of the section. Loops, arrays, ArrayLists and 2D arrays are where most points live. Unit 3 is the smallest slice here, but it gets a whole free-response question of its own.

Strategies that work on this section

- **Build a trace table.** One column per variable, one row per loop pass. Don't do it in your head.
- **Check the loop bounds first.** `<` vs `<=` and starting at 0 vs 1 decide most answers.
- **Watch integer division.** `7 / 2` is `3`, not 3.5.
- **Plug in a tiny example** for "which code works for all cases" questions. Try an empty list, one element, and a negative number.
- **Flag and move on** after 3 minutes. Come back once the easy points are banked.
- **Use the Java Quick Reference** to confirm method names and return types.

With about **2 minutes** each, a 6-pass loop trace is fine. A 50-pass loop means there's a pattern to spot instead!



03 WHAT THE COURSE COVERS

The 4 units, in plain English

Everything on the exam fits into these four units.

UNIT

1

15–25%

Using Objects & Methods

Variables, data types, expressions, and calling methods on objects someone else wrote.

Be ready to: use `int`, `double`, `boolean`; cast and spot integer division; use `%`; call static and instance methods; use the `Math` class; create objects with `new`; work with `String` methods such as `substring`, `indexOf`, `equals` and `compareTo`.

UNIT

2

25–35%

Selection & Iteration

Making decisions and repeating work: if statements, loops and the classic loop algorithms.

Be ready to: write and simplify boolean expressions (including De Morgan's laws and short-circuiting); use `if / else if`; write `while` and `for` loops; nest loops; loop through Strings; count loop iterations; build sum, min/max, count and digit-extraction algorithms.

UNIT

3

10–18%

Class Creation

Designing your own classes: what objects store and what they can do.

Be ready to: write a class header, `private` instance variables and constructors; write accessor, mutator and other methods; explain `static` variables and methods; reason about scope and `this`; pass and return object references (and spot aliasing).

UNIT

4

30–40%

Data Collections

Storing and processing lots of data. The biggest unit and the heart of the exam.

Be ready to: create and traverse arrays, `ArrayList`s and 2D arrays (row-major and column-major); use wrapper classes; read data from text files; run standard algorithms; trace linear and binary search; trace selection, insertion and merge sort; trace recursive methods; discuss data privacy and bias.

Using old review books? The current course no longer includes **inheritance** (superclasses, subclasses, `super`, polymorphism). Skip those chapters, and spend that time on `ArrayList` and 2D array practice.



New to coding? Many students take CSA after AP CSP. The big jump is writing exact Java syntax yourself, so start typing small programs in week one.

04 SECTION II The 4 free-response questions

90 minutes, 4 questions, 25 points. The question types never change, so you can practice each one on purpose. Every question tests your ability to write working Java code.

Q1 · 7 pts

Q2 · 7 pts

Q3 · 5 pts

Q4 · 6 pts

25 raw points in total, worth 45% of your AP score

Q1

Methods & Control Structures

7 points · Part A 4 + Part B 3

Write two methods (or a constructor and a method) for a class you're given. **Part A** needs loops and/or if statements plus calls to the class's own methods. **Part B** needs **String methods**.

Practice: `substring` boundaries, `indexOf` returning -1, building a String in a loop.

Q2

Class Design

7 points

Write a complete class from a description and a table of example calls. It must have a class header, instance variables, a constructor and the required method(s). A second class may be involved.

Practice: turning an example table into instance variables and exact method headers.

Q3

Data Analysis with ArrayList

5 points

Write one method that uses, analyzes or changes data stored in an **ArrayList**, usually a list of objects you access through getter methods.

Practice: `get`, `size`, `add`, `remove`, and removing safely while looping.

Q4

2D Array

6 points

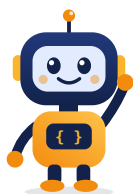
Write one method that uses, analyzes or changes data in a **2D array**, such as finding a row or column that meets a condition, or counting neighbors.

Practice: `grid.length` (rows) vs `grid[0].length` (columns), column-major loops.

• A time plan for the 90 minutes

Skim 3 min	Q1 ~20 min	Q2 ~22 min	Q3 ~19 min	Q4 ~20 min	Check 6 min
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A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Rubric points are given for **individual correct pieces**: the loop, the condition, the return. A half-finished method can still earn most of the points!

05 Anatomy of a full-credit answer

FREE-RESPONSE PLAYBOOK

A short Class Design practice task (Q2 style), a clean answer, and the habits that earn points.

PRACTICE TASK · WRITTEN BY ACT SAT GO

Write the **WaterBottle** class. A bottle is created with a capacity in milliliters and starts empty. `fill(ml)` adds water but never goes past the capacity. `drink(ml)` removes up to `ml` of water and returns how much was actually drunk.

Statement	Value returned	Explanation
<code>WaterBottle w = new WaterBottle(500);</code>	–	Capacity 500, holds 0
<code>w.fill(300); w.fill(400);</code>	–	Holds 500, not 700
<code>w.drink(200);</code>	200	Holds 300
<code>w.drink(900);</code>	300	Only 300 left, now empty

```

public class WaterBottle ①
{
    private int capacity; ②
    private int amount;

    public WaterBottle(int cap) ③
    {
        capacity = cap;
        amount = 0;
    }

    public void fill(int ml) ④
    {
        amount = Math.min(capacity,
                           amount + ml);
    }

    public int drink(int ml)
    {
        int drunk = Math.min(ml, amount);
        amount -= drunk;
        return drunk; ⑤
    }
}

```

- ① **Class header** spelled exactly as given: `public class WaterBottle`.
- ② **Private instance variables** for everything the object must remember between calls.
- ③ **Constructor** with the right parameter. Initialize every instance variable. Don't redeclare them (`int capacity = cap;` creates a new local variable and loses the value).
- ④ **Method headers** match the examples: name, parameter type and return type (`void` vs `int`).
- ⑤ **Return, don't print.** If a method should give back a value, use `return`, not `System.out.println`.

Five habits for every free-response question

- 1 Read the examples table before writing anything.
- 2 Use methods the question gives you. Don't rewrite them.
- 3 Trace your code with one example from the question.
- 4 No imports, comments or `main` needed.
- 5 Stuck? Write the header and the loop anyway.

LEARN FROM OTHERS

06

10 Java traps that cost points

Meet **Glitch**. He lives inside these ten mistakes. Squash each one before May.

Off by one? **That's my favorite!**



01 Integer division

`7 / 2` is `3`. Cast first if you need a decimal:
(double) `7 / 2` is `3.5`.

02 Comparing Strings with ==

`==` checks if two references point to the same object. Use `s1.equals(s2)` for content.

03 Looping one step too far

`i <= arr.length` crashes with `ArrayIndexOutOfBoundsException`. Last index is `length - 1`.

04 Removing while looping forward

After `list.remove(i)`, everything shifts left and the next item is skipped. Loop backwards instead.

05 length, length() or size()?

Arrays: `arr.length`. Strings: `s.length()`.
ArrayLists: `list.size()`.

06 substring's end is exclusive

`"HELLO".substring(1, 3)` is `"EL"`: it stops *before* index 3.

07 Rows and columns swapped

`grid.length` is the number of rows;
`grid[0].length` is the number of columns.

08 Redeclaring in the constructor

`int count = start;` makes a new local variable. Write `count = start;`.

09 Printing instead of returning

If the method's header says `int` or `boolean`, it must `return` a value on every path.

10 Leaving a free-response blank

Points are given for each correct piece. A header plus a sensible loop can still score.

Quick self-check: could you explain, without running it, why this prints `2.0` and not `2.5`? If yes, trap 1 can't catch you.

```
int a = 5, b = 2;
double r = a / b;
System.out.println(r);
```

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, but keep the order: objects and loops first, then classes, then data collections.



Sep – Nov
Foundations



Units 1 & 2

≈ half of MCQ

Types, casting, String methods, if statements, while and for loops. Trace 5 short code snippets a week and write one small program each weekend.

Dec – Jan
Build classes



Unit 3 + your first FRQs

Q1 & Q2

Write classes with constructors, getters and methods. Do one Class Design and one Methods & Control Structures practice question every week.

Feb – Mar
Data collections



Unit 4

biggest unit

Arrays, ArrayLists, 2D arrays, searching, sorting and recursion. Add ArrayList and 2D Array free-response practice (Q3 & Q4) as you finish each topic.

April
Full practice



Timed practice in Bluebook

Open the Bluebook test preview. Take one full 42-question set and one full 4-question free-response set under real timing. Log every mistake by unit.

Early May
Sharpen



Fix weak spots

Redo every question you missed. Review the Java traps and the Quick Reference methods.

Exam day
May



AP CSA exam · Bluebook

Sleep well, charge your device, and use the exam-day checklist at the end of this guide.

◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: one timed set of 21 multiple-choice questions plus 2 free-response questions	List your two weakest units
3 weeks out	Fix your weakest unit; one free-response question of each type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam in Bluebook conditions	Finish both sections on time
Final week	Light review: traps list, String and ArrayList methods, 2D loops	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **20 minutes** for all ten, then check page 11.

1 UNIT 1 · EXPRESSIONS

```
int a = 17;
int b = 5;
double c = a / b + a % b;
System.out.println(c);
```

What is printed?

- A 5.0
- B 5.4
- C 3.4
- D 7.0

2 UNIT 1 · STRING METHODS

```
String s = "PROGRAMMING";
System.out.println(
    s.substring(3, 7) + s.indexOf("M"));
```

What is printed?

- A GRAMM6
- B GRAM6
- C GRAM7
- D OGRA6

3 UNIT 2 · LOOPS

```
int count = 0;
for (int i = 1; i <= 20; i += 3)
{
    if (i % 2 == 0) { count++; }
}
```

What is the value of `count` after this code runs?

- A 3
- B 4
- C 6
- D 7

4 UNIT 2 · BOOLEAN LOGIC

Which expression is equivalent to the one below for all integers `x` and `y`?

```
!(x > 5 || y <= 2)
```

- A `x <= 5 || y > 2`
- B `x < 5 && y > 2`
- C `x > 5 && y <= 2`
- D `x <= 5 && y > 2`

5 UNIT 2 · WHILE LOOPS

Consider the code segment below.

```
int n = 1234;
int sum = 0;
while (n > 0)
{
    sum += n % 10;
    n /= 10;
}
System.out.println(sum);
```

What is printed when the code segment runs?

- A 4
- B 24
- C 10
- D 1234

08 TEST YOURSELF (continued)

6 UNIT 3 · OBJECTS & REFERENCES

```
public class Counter
{
    private int count;
    public Counter(int start) { count = start; }
    public void increment() { count++; }
    public int getCount() { return count; }
}
```

Consider the class on the left. What is printed when this code runs?

```
Counter c1 = new Counter(5);
Counter c2 = c1;
c2.increment();
c1.increment();
System.out.println(c1.getCount()
    + " " + c2.getCount());
```

- A 6 6 B 6 7
C 7 7 D 5 7

7 UNIT 4 · ARRAYLIST

`nums` is an `ArrayList<Integer>` containing `[4, 8, 8, 3, 8]`. What does it contain after this code runs?

```
for (int i = 0; i < nums.size(); i++)
{
    if (nums.get(i) > 5)
    {
        nums.remove(i);
    }
}
```

- A [4, 3]
B [4, 8, 3]
C [4, 8, 3, 8]
D An `IndexOutOfBoundsException` occurs

8 UNIT 4 · 2D ARRAYS

What is the value of `sum` after this code runs?

```
int[][] grid = { {1, 2, 3},
                 {4, 5, 6},
                 {7, 8, 9} };
int sum = 0;
for (int r = 0; r < grid.length; r++)
{
    sum += grid[r][grid.length - 1 - r];
}
```

- A 12 B 45
C 18 D 15

9 UNIT 4 · RECURSION

What value is returned by `mystery(7)`?

```
public static int mystery(int n)
{
    if (n <= 1)
    {
        return 1;
    }
    return n + mystery(n - 2);
}
```

- A 15 B 28
C 16 D 13

10 UNIT 4 · SORTING

The array `{5, 2, 9, 1, 7}` is sorted into ascending order using **selection sort**. What does the array look like after the first **two** passes of the outer loop?

- A {1, 2, 9, 5, 7}
B {1, 2, 5, 9, 7}
C {2, 5, 9, 1, 7}
D {1, 2, 5, 7, 9}

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	A · 5.0	<code>17 / 5</code> is integer division, so it's 3. <code>17 % 5</code> is 2. <code>3 + 2 = 5</code> , and storing it in a <code>double</code> prints <code>5.0</code> . Choice B forgets that integer division drops the decimal.
2	B · GRAM6	Indexes start at 0. <code>substring(3, 7)</code> takes indexes 3–6: "GRAM" (index 7 is excluded). The first "M" is at index 6.
3	A · 3	<code>i</code> takes the values 1, 4, 7, 10, 13, 16, 19. Only 4, 10 and 16 are even. Choice D counts every pass.
4	D	De Morgan's law: <code>!(A B)</code> becomes <code>!A && !B</code> . The opposite of <code>x > 5</code> is <code>x <= 5</code> , and the opposite of <code>y <= 2</code> is <code>y > 2</code> .
5	C · 10	<code>n % 10</code> gets the last digit and <code>n /= 10</code> removes it: <code>4 + 3 + 2 + 1 = 10</code> . This digit-sum pattern is a classic.
6	C · 77	<code>c2 = c1</code> copies the <i>reference</i> , not the object. Both variables point to one Counter, which is incremented twice: <code>5 → 7</code> . This is called aliasing.
7	B · [4, 8, 3]	When index 1 is removed, the second 8 slides into index 1, but <code>i</code> moves on to 2 and skips it. Fix: loop from the end, or only increase <code>i</code> when nothing is removed.
8	D · 15	The column index runs 2, 1, 0 as the row runs 0, 1, 2. That is the anti-diagonal: <code>3 + 5 + 7 = 15</code> .
9	C · 16	<code>mystery(7) = 7 + mystery(5) = 7 + 5 + mystery(3) = 7 + 5 + 3 + mystery(1)</code> . The base case returns 1, so the total is 16.
10	A	Pass 1: smallest value 1 swaps with 5 → {1, 2, 9, 5, 7}. Pass 2: smallest remaining value 2 is already in place, so nothing changes. Choice C is what insertion sort gives after two passes.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Start with tracing loops and integer math (Units 1–2).

10 KNOW THE TARGET How students score

In recent years, about 66–67% of students earned a 3 or higher on AP Computer Science A, and about 25–26% earned a 5. CSA scores are spread wide, so preparation makes a real difference.

66–67%

of students earned a 3 or higher in recent years

25–26%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, with the charger packed
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ No calculator needed for this exam
- ☐ Plan your time: **just over 2 min** per multiple-choice question; about **20 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Keep the **Java Quick Reference** open when you're unsure of a method
- ☐ Write **something** for every free-response part, starting with the method header

In the free-response section: match the method headers exactly, use the methods the question gives you, return values instead of printing, and trace one of the given examples through your code before moving on.



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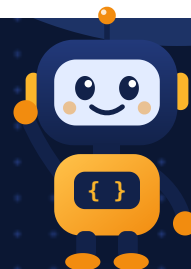
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Book a free diagnostic lesson. We'll find your weakest units, review your free-response code line by line, and give you a week-by-week plan to exam day in May.

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Live 1-on-1 sessions on all four units, with tracing drills after every lesson.

Free-response coaching

Practice all four question types and get your code reviewed against rubric-style criteria.

Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

◆ Contact us

Web www.actsatgo.com

Email info@actsatgo.com

Phone [+1 \(945\) 391-6179](tel:+19453916179)

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